VANSHIKA SOHAL

B.Tech CSE | Thapar Institute of Engineering and Technology, Patiala (2024–2028)
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GitHub: Github · LinkedIn: LinkedIn · Leetcode: Leetcode

Data Structures & Algorithms: • Solved ~240 problems on LeetCode

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About Me

A self-driven and enthusiastic Computer Science student passionate about building impactful projects, exploring Machine Learning, and contributing to collaborative team environments. Adept at problem-solving, web development, and data structures with a strong foundation in C++, Python, and Flask.

Skills

- Languages & Programming: C, C++, Python
- Frontend Development: HTML, CSS, React.js, JavaScript
- Backend Development & Database: Flask, SQLite,FastApi
- Machine Learning: Supervised & Unsupervised Learning (Scikit-learn, Pandas, NumPy, Matplotlib)
- Core CS Concepts: Data Structures & Algorithms (DSA)
- Tools & Technologies: Git, GitHub, VS Code

Projects

Sorting Algorithm Benchmark | GitHub

- Compared performance of Bubble Sort, Merge Sort, and Quick Sort using Python.
- Measured execution time across multiple array sizes and visualized results with Matplotlib.
- Demonstrated understanding of algorithm efficiency and runtime analysis.

AYUSH FHIR Terminology Portal – SIH 2025 | GitHub

- Contributing as Backend Developer for a healthcare portal bridging AYUSH with modern healthcare standards.
- Developed FastAPI modules for search, translation, and AI symptom analysis.
- Assisted in data pipelines and deployment with FHIR R4 compliance.
- Tech Stack: FastAPI, Python, FHIR, AI modules

EMI & Loan Prediction | GitHub

- Full-stack ML web application predicting loan eligibility using classification & regression models.
- Built frontend with React and backend with Flask; deployed for real-time user interaction.
- Dataset sourced from Kaggle.
- Tech Stack: React.js, Flask, Python (Scikit-learn, Pandas, NumPy), SQLite

Autonomous AI Email Agent (n8n + Google Gemini) | GitHub

- Built an autonomous AI agent capable of understanding natural language and sending emails automatically.
- Implemented conversational memory for context-aware responses.
- Deployed via n8n.io workflow for easy integration and rapid deployment.
- Tech Stack: n8n.io, Google Gemini, Gmail API

Achievements

- Cleared SIH Internal Round 2025
- Selected for Unstop Mentorship Program
- GSSOC 2025 Contributor
- **Best Management Skills Award** *Shoonya 5.0 (2025)* – TMS Thapar Society
  • Coordinated event operations, managing teams and timelines